

USING STATA® TO IDENTIFY

The Relationship Between PM2.5 Air Pollution And A Country's Child Mortality

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1. Introduction

In February 2024, the severe air pollution in the city of Milan sparked a widespread discussion and concerns about the city's environmental issues. During that time, Milan was ranked among the most polluted cities in the world by the Swiss air quality technology company IQAir, with record-high PM2.5 levels detected by citizens and air quality monitoring agencies in this northern Italian economic capital of fashion, design and industries.

This recent event prompted me to choose the study topic on the link between PM2.5 pollution and child mortality rate, as an assignment for the course Applied Environmental and Resource Economics, using the statistical software STATA. Although there is comprehensive quantified research in the literature on this topic, the main goal of this study is to understand the research context, learn more about the empirical strategy, and interpret key findings.

According to the European Environment Agency (EEA) air quality reporting annual statistics (Figure 1), Italy suffers air pollution level higher than most EU member states, with annual PM2.5 exposure mostly above EU standards and at least 4 times above WHO standard. Figure 1 also shows that the region in which Milan is located reports the highest concentration of PM2.5.

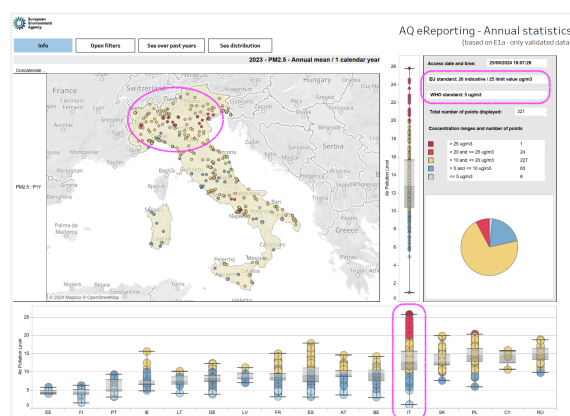


Figure 1. Air quality statistics by the European Environment Agency (EEA). 2023

Air pollution is recognized as one of the main risk factors for Non-Communicable Diseases by WHO. It's one of the most significant environmental risks to health in Europe: 569,000 premature deaths can be attributed to ambient air pollution in 2019 ^[1]. The EEA deemed PM2.5 to cause the most damaging impacts on human health ^[2]. Italy has the largest number of premature deaths attributable to long-term exposure to PM2.5 per year (55,813 on average across the year 2005 to 2020), followed by Germany (52,048) and Poland (43,472) ^[3].

2. Variables

2.1. Dependent Variable - Child Mortality Rate

The dependent variable in this study is child mortality rate, specifically the under-five (per 1,000 live births) mortality rate. Under-five mortality rate is the probability per 1,000 that a newborn baby will die before reaching age five, if subject to age-specific mortality rates of the specified year. This study uses the estimates developed by the UN Inter-agency Group for Child Mortality Estimation.

2.2. Independent Variable - PM2.5 Pollution

The independent variable of interest in this study is the PM2.5 air pollution, specifically the population-weighted exposure to ambient PM2.5 pollution ($\mu\text{g}/\text{m}^3$) which is defined as the average level of exposure of a nation's population to concentrations of suspended particles measuring less than 2.5 microns in aerodynamic diameter. Exposure is calculated by weighting mean annual concentrations of PM2.5 by population in both urban and rural areas. This study sources the PM2.5 data from Global Health Data Exchange published by Institute for Health Metrics and Evaluation 2021 ^[4].

Based on the basic analysis of the two main variables in [Section 4.1](#), I suspect their relationship to evolve over time due to the advancement of technologies and healthcare quality. Therefore, a dataset of GDP per capita (GDP_{pc}) from World Bank Fiscal Year 2024 is also included for this analysis. In addition to that, all 3 raw datasets used in this study include a variable named "Income Group" which is clearly defined by the World Bank Group as shown in Figure 2.

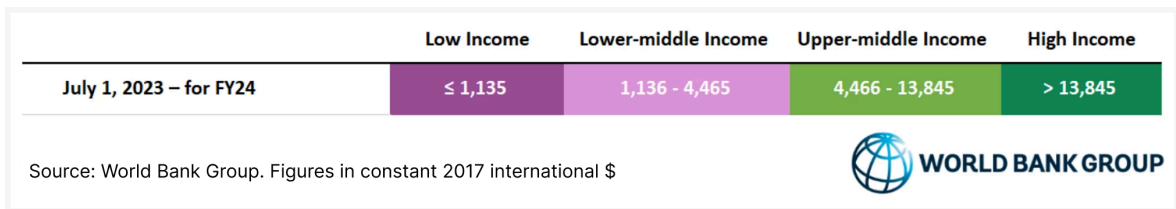


Figure 2. Work Bank Income Group Classification

3. Econometric Model

3.1. Dataset Characteristics

The child mortality rate and PM2.5 datasets provide these variables: Country Name, Country Code, Indicator Name, Indicator Code, Region, Income Group (comprised of 4 groups from high income to low income), Year (1960-2022, although there are missing data from some years and countries). The data are formatted in the wide format in Excel files. After a basic examination on Excel, the characteristics of the datasets can be summarized as below:

- Time variation: the data sets include observations over time (years).
- Country-specific: the data sets provide observations in 266 entities that consist of 193 sovereign countries, 23 dependent territories, and other various groupings. Those groupings are considered not relevant and therefore have been filtered out. So the data analyses onwards only deal with the 216 countries and their territories.

As Figure 3a shows, the world child mortality rate has an obvious declining trend over the years, while Figure 3b and 3c show that the PM2.5 data have a more complex trend over the years worldwide, with missing data before 1990.

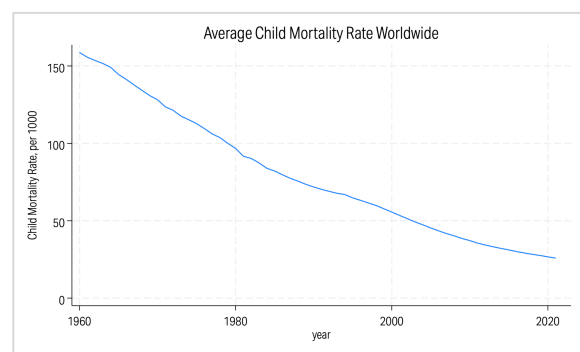


Figure 3a. Average child mortality rate worldwide.

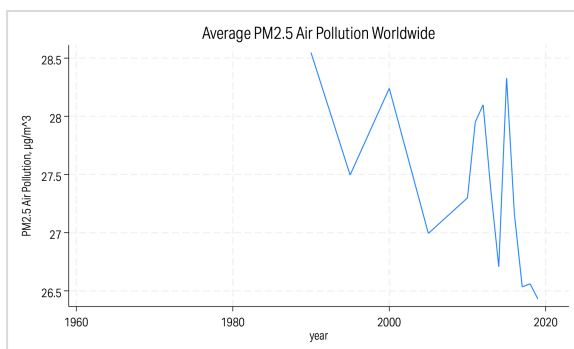


Figure 3b. Average PM2.5 exposure worldwide since 1960.

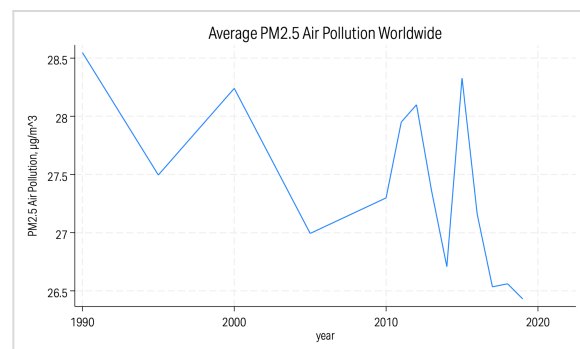


Figure 3c. Average PM2.5 exposure worldwide since 1990.

3.2. Choosing A Panel Data Model

This assignment aims to use statistical methods to find the empirical relationship between child mortality rate (a social and economic outcome) and PM2.5 air pollution

(one of the weather/climate variables). Among the 3 common climate econometrics (cross-sectional, panel data, and long-difference models), I choose to apply the panel data approach, given the dataset characteristics mentioned in [Section 3.1](#). That is because the panel data model allows me to control for country-specific effects as well as year fixed effects.

3.3. Main Equations

Although the climate impact on human behavior (health, mortality, consumption, migration, labor, etc.) is complex, it's useful to start from a simple econometric model and advance with the findings along the way. This study first explores the linear relationship between the two main variables, then assumes a quadratic relationship between them, as a number of studies have found evidence of non-linear relationships between climate variables (e.g. temperature, CO₂ emission) and socioeconomic outcomes (e.g. crime, mortality) in France, China, South and East Asia, Latin America and the Caribbean countries [\[5-9\]](#).

Therefore, this study uses these equations below for the regression analysis:

$$y_{it} = \alpha_{it} + \beta_1 x_{it} + \varepsilon_{it} \quad (1)$$

$$y_{it} = \alpha_{it} + \beta_1 x_{it} + \gamma_1 z_{it} + \varepsilon_{it} \quad (2)$$

$$y_{it} = \alpha_{it} + \beta_1 x_{it} + \beta_2 x_{it}^2 + \varepsilon_{it} \quad (3)$$

$$y_{it} = \alpha_{it} + \beta_1 x_{it} + \beta_2 x_{it}^2 + \gamma_1 z_{it} + \varepsilon_{it} \quad (4)$$

$$y_{it} = \alpha_{it} + \beta_1 x_{it} + \beta_2 x_{it}^2 + \gamma_1 z_{it} + \gamma_2 z_{it}^2 + \varepsilon_{it} \quad (5)$$

Where:

- y_{it} represents the child mortality rate of country i in year t , which is the under-5 mortality rate (per 1,000 live births).
- x_{it} represents the PM2.5 air pollution of country i in year t , which is the mean annual exposure, $\mu\text{g}/\text{m}^3$. The coefficients β_1 and β_2 are the marginal effect corresponding to the linear term x_{it} and the quadratic term x_{it}^2 respectively.
- z_{it} represents the GDP per capita of country i in year t , which is GDP per capita based on purchasing power parity (constant 2017 international \$). The coefficients γ_1 and γ_2 are the marginal effect corresponding to the linear term z_{it} and the quadratic term z_{it}^2 respectively.

- α_{it} represents the country-time-specific intercept, which captures unobserved heterogeneity that varies both across countries and over time.
- ε_{it} is the error term.

In the attached panel.do file, models are named after each equation listed here, e.g. Model 1, Model 5, etc.

4. Results and Discussion

4.1. Trends of Child Mortality Rate & PM2.5 Pollution Across Countries

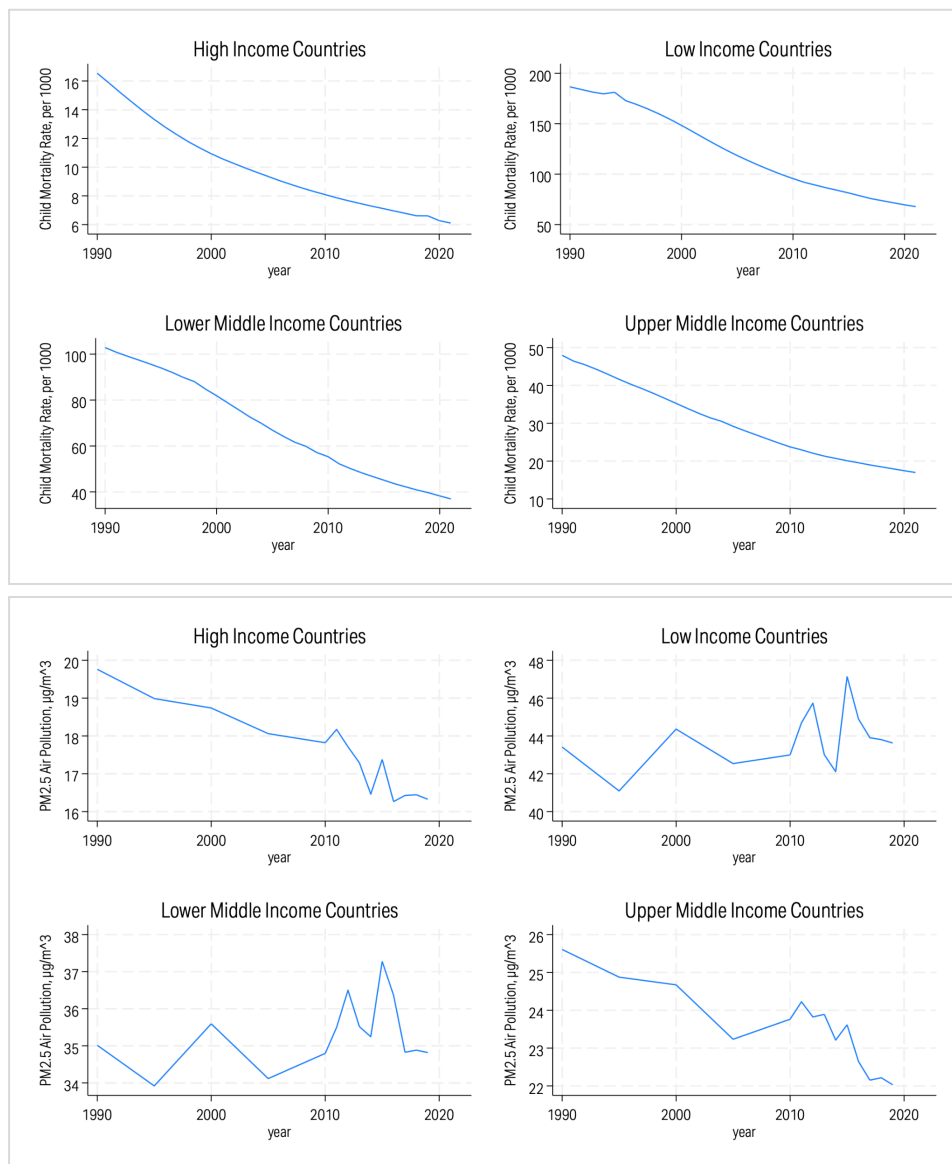


Figure 4a. Annual mean child mortality rate by income level of the country.

Figure 4b. Annual mean PM2.5 exposure by income level of the country.

Figure 4a indicates that despite the different child mortality rate levels among the countries, all income groups' rates have been decreasing over the last 30 years. Meanwhile the annual average PM2.5 exposure has only declined in the higher-income country groups (Figure 4b).

4.2. Comparing Models With & Without Controlling for GDP_{pc}

Based on the Environmental Kuznets Curve (EKC) hypothesis, I decide to use the quadratic terms of the independent variables and run Model 1 to Model 5 with the *reghdfe* command (high-dimensional regressions with fixed effects) in STATA. The results are shown in Table 1.

	(1)	(2)	(3)	(4)	(5)
	mort	mort	mort	mort	mort
pm	-1.41818*** (0.22781)	-1.11999*** (0.21003)	-3.28247*** (0.63347)	-2.46857*** (0.59561)	-2.41567*** (0.59316)
gdp		0.00106*** (0.00032)		0.00098*** (0.00032)	0.00205*** (0.00057)
pm_sq			0.02047*** (0.00652)	0.01437** (0.00595)	0.01434** (0.00597)
gdp_sq					-0.00000** (0.00000)
R ² (within)	0.89205	0.90394	0.89440	0.90514	0.90715
Obs.	2688.00000	2523.00000	2688.00000	2523.00000	2523.00000

Table 1. Regression results of Model 1-5. Notes: * p<0.10 ** p<0.05 *** p<0.01 Standard errors in parentheses. Abbreviations of variables in the table represent: **mort**: child mortality rate; **pm**: PM2.5 exposure; **pm_sq**: the squared value of PM2.5 exposure; **gdp**: GDP per capita; **gdp_sq**: the squared value of GDP per capita.

Overall the proposed models fit the data well, with more than 89% of the observations explained by the independent variables and all coefficients statistically significant. Models that include GDP_{pc} (higher Within R²) explain within-group variations better than those excluding GDP_{pc}.

Contrary to common beliefs, all models imply that PM2.5 has a negative impact on child mortality, while GDP_{pc} has a positive impact on child mortality. The greater magnitude of coefficients in Model 3-5 also suggest that the relationship between PM2.5 and child mortality is more likely to be non-linear rather than linear: a quadratic one is sufficient and plausible.

Comparing Model 1 vs 2, Model 3 vs 4, and Model 3 vs 5, the absolute values of the PM2.5 coefficients decrease (21%, 24.8%, 26.4%) whenever GDP_{pc} is introduced. This means PM2.5's influence on child mortality is overestimated without controlling for GDP_{pc}. It's now evident that GDP_{pc} should not be omitted, without which my estimates would be biased. Given the coefficient of *gdp_sq* equals 0 in Model 5, I choose Model 4 to proceed with further analysis.

4.3. Predicted Response of Child Mortality Rate to PM2.5

Plotting Model 4's calculations, it shows that overall, taking all countries into account, child mortality rate responds to PM2.5 negatively: it drops as PM2.5 exposure increases, then continues to fall slowly, even below 0 at about $40\mu\text{g}/\text{m}^3$ (Figure 5a). This is not realistic as mortality rate can't be negative.

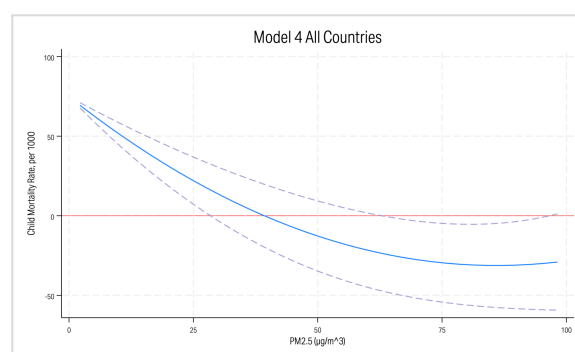


Figure 5a. Predicted response of child mortality rate to PM2.5.

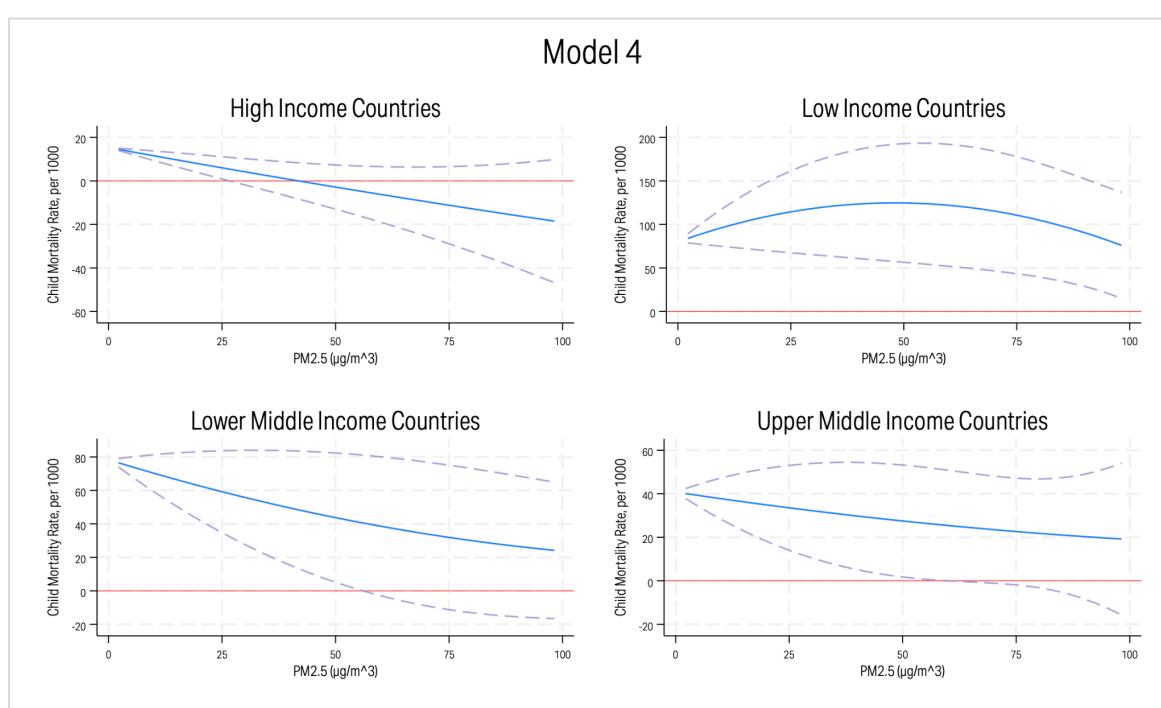


Figure 5b. Predicted response of child mortality rate to PM2.5 in each income group.

Dash lines represent the upper & lower 90% confidence intervals.

However, if we look into each income group (Figure 5b), this global (almost linear) declining pattern of child mortality rate as PM2.5 pollution worsens does not represent the one in low income countries, where it shows a reverse-U-shape corresponding to the EKC curve. Figure 5a & 5b imply that the explaining power of GDP_{pc} might be overpowering and thus masking the true effects of PM2.5. The wide confidence intervals also indicate a lot of uncertainty in the predictions.

4.4. Marginal Effects of PM2.5 on Child Mortality Rate

From another perspective, I plot the marginal effects of PM2.5 on child mortality rate using the quadratic term of both PM2.5, GDP_{pcr} , and the interaction between PM2.5 and GDP_{pc} . The result is somewhat similar to that in Model 4: negative correlation but child mortality falls at a faster rate as PM2.5 pollution reaches higher levels (Figure 6a).

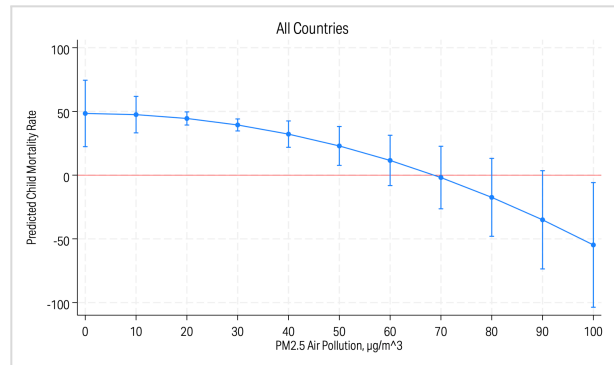


Figure 6a. Marginal effects of PM2.5 on child mortality rate.

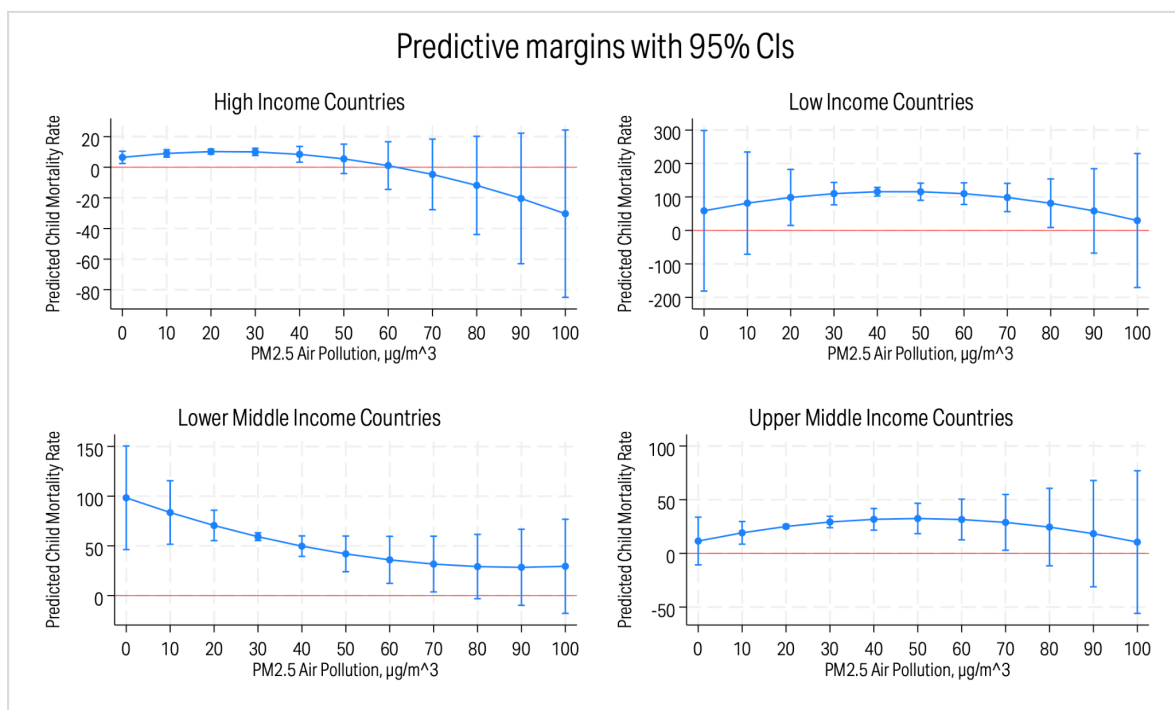


Figure 6b. Predictive marginal effects of PM2.5 on child mortality rate in each income group.

When analyzing each income group (Figure 6b), we see curves similar (though with a much milder curvature) to the EKC's, except in the lower-middle income group, which shows a strong negative correlation between PM2.5 and child mortality. Overall, the confidence of these predicted margins is weak. The only certain thing is: high income countries appear to have very low child mortality rates regardless of PM2.5 pollution levels.

4.5. Logarithm Transformation to Address Right-Skewness

The results till this point are somehow confusing and therefore unsatisfactory. So I'm inclined to explore a new way of visualizing the data.

It turns out all 3 sets of data have right-skewed distribution (Figure 7). So it's reasonable to consider logarithm transformation of the variables. After applying natural logarithm $\ln(x)$, we have 5 new log-log models as displayed in Table 2 below:

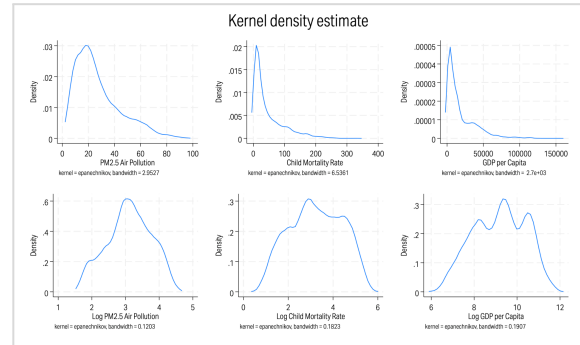


Figure 7. Right-skewed distribution of the variables.

	(1b)	(2b)	(3b)	(4b)	(5b)
	log_mort	log_mort	log_mort	log_mort	log_mort
log_pm	0.08192 (0.10461)	0.01511 (0.08894)	0.63449 (0.48055)	0.45550 (0.41778)	0.48958 (0.42254)
log_gdp		-0.27526*** (0.07023)		-0.27541*** (0.06934)	-0.40213 (0.36755)
log_pm_sq			-0.09225 (0.07361)	-0.07333 (0.06487)	-0.07695 (0.06473)
log_gdp_sq					0.00731 (0.02064)
R ² (within)	0.98081	0.98385	0.98088	0.98390	0.98391
Obs.	2688.00000	2523.00000	2688.00000	2523.00000	2523.00000

Table 2. Regression results of Log-log Model 1b-5b. Notes: * p<0.10 ** p<0.05 *** p<0.01 Standard errors in parentheses.

Abbreviations of variables in the table represent: **log_mort**: natural logarithm of child mortality rate; **log_pm**: natural logarithm of PM2.5 exposure; **log_pm_sq**: the squared value of log_pm; **log_gdp**: natural logarithm of GDP per capita; **log_gdp_sq**: the squared value of log_gdp.

None of the log PM2.5 coefficients are statistically significant in the log-log models. Only the log GDP_{pc} in Model 2b (linear) and 4b (log PM2.5 quadratic term, log GDP_{pc} linear term) exhibits significant negative effect on log child mortality. In these models, log PM2.5 might have a strong positive non-linear impact on log child mortality even though it's not statistically significant. The within R² values of the log-log models (0.98+) are all greater than those of non-log models (0.89~0.90, c.f. Table 1), which means the log models explains the data much better.

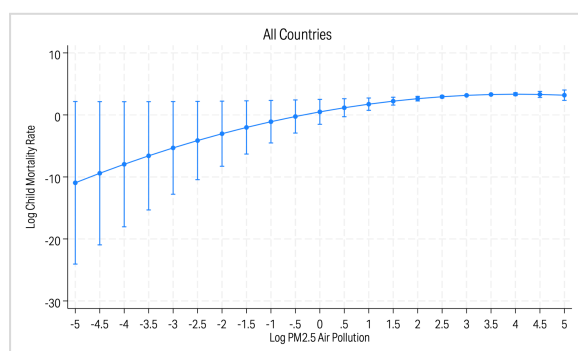


Figure 8a. Predictive marginal effects of log PM2.5 on log child mortality rate.

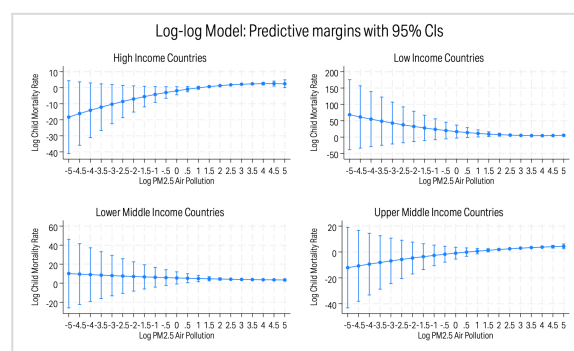


Figure 8b. Predictive marginal effects of log PM2.5 on log child mortality rate in each income group.

The plots in Figure 8 picture a scenario where PM2.5 pollution ranges from 0.007 $\mu\text{g}/\text{m}^3$ to 150 $\mu\text{g}/\text{m}^3$. Log child mortality rate in lower-income groups either decreases as log PM2.5 increases or remains almost insensitive. In contrast, in higher-income groups, log child mortality rate follows an upward path of a reverse-U-shape curve reaching a plateau as PM2.5 pollution becomes more severe.

5. Conclusion

The relationship between child mortality and PM2.5 air pollution is not linear; the assumption of a quadratic relationship is plausible. It's biased to conclude PM2.5's negative or positive effect on child mortality without considering the development level of each country (e.g. including GDP_{pc} as a continuous variable, or including income groups as a category). Child mortality rates in poor countries and rich countries exhibit distinctive responses to PM2.5.

However, it's worth mentioning that the interactions between PM2.5 emissions, industrial activities, environmental policies, and GDPs are complex. The predicted effects of PM2.5 in this study might be overpowered or masked by any of these factors. We know there is some level of PM2.5 impact on child mortality but its true nature remains uncertain and requires further analysis, e.g. other types of non-linear models, a structural econometric approach, etc. This study has not taken into account other potential macro influencers such as public healthcare expenses or temperature, thus the results should be interpreted in a cautious manner.

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